HACKATHON SUBMISSION · AI FOR BHARAT & PUBLIC SERVICES

Bharat Setu · भारत सेतु

An AI bridge between Indian citizens and the public services they are entitled to — multilingual, explainable, offline-first.

43 schemes | 30 categories | 83 documents | 12 helplines | 13 languages | 63/63 tests | ₹0 to the citizen

Live: webapp-kohl-six-95.vercel.app | Source: github.com/no6546559-sketch/bharat-setu

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Project Report — AI for Bharat & Public Services

Submission for: Hackathon — theme *AI for Bharat & Public Services* **Live application:** https://webapp-kohl-six-95.vercel.app **Source code:** https://github.com/no6546559-sketch/bharat-setu **Team size:** solo–4 (project is designed for a small team; all roles merged in this build) **Report date:** 11 October 2026

1. Executive Summary

India operates one of the largest social-welfare systems on earth. Yet a large share of eligible citizens never receive the benefits they are legally entitled to. The binding constraint is almost never money — it is **information asymmetry**: schemes are scattered across dozens of central and state portals, most interfaces are English-first, the forms are not explained, and a layer of middlemen profits from that confusion.

Bharat Setu ("India Bridge") is a **multilingual, explainable, offline-first AI assistant** that closes that gap. A citizen enters a short profile once and receives:

- every government scheme they are eligible for, **with the exact reason** they qualify;
- a **document-readiness score** telling them precisely what is missing and where to get it;

- a **formal RTI / grievance / certificate letter** they can print and submit the same day;
- an estimated **annual benefit value in rupees**, tracked application-by-application;
- a **What-if simulator** that turns eligibility into a live dial — move age and income and watch

what opens up or closes, without touching the saved profile;

- a **Household plan** that combines several family members into one benefit total;
- a **fraud shield** that reads a suspicious SMS and explains, in plain language, exactly

what is happening and what to do next.

It does all of this in **13 Indian languages** (including RTL Urdu), by **voice or text**, and **with no internet connection required** — because the entire reasoning engine runs on the device.

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2. Problem Statement

2.1 The gap

Barrier	Consequence for the citizen
Fragmented discovery — schemes live on dozens of separate portals	Eligible people never learn they qualify
Language — most portals are English-first	Crores of citizens are locked out of the interface
Opaque eligibility rules	People don't know *why* they were rejected or what to fix
Documentation chaos	Applications stall on a single missing certificate
No recourse knowledge	Citizens don't know RTI or grievance mechanisms exist
Fraud and middlemen	Fake "benefit" SMS, processing-fee scams, touts

2.2 Who suffers most

First-time internet users, rural households, women, senior citizens, persons with disabilities, and low-literacy users — precisely the groups the welfare system is meant to protect. Any solution must therefore be **simple, spoken, localised and trustworthy**, not merely accurate.

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3. Solution Overview

Bharat Setu is a **single installable web app** with eight tightly integrated views, plus three embedded decision tools (What-if simulator, Household mode, benefit estimator):

#	View	What it does
1	🏠 **Home**	Explains the product, one-tap entry points, key numbers
2	🧭 **Scheme Finder**	Profile → ranked eligibility list **with reasons and blockers**; live **₹ value pill**; **What-if simulator**; **Household mode**
3	📋 **My Plan**	Application tracker + document bar + **₹ benefit estimator**
4	🤖 **AI Assistant**	Grounded Q&A in 13 languages, voice in and out
5	📄 **Document Checker**	Readiness score, missing docs, issuing office + timeline
6	✍️ **RTI & Letters**	RTI, grievance, scheme representation, certificate request
7	🛡️ **Fraud Shield**	Risk score /100 + red flags + exact next actions
8	📞 **Helplines & Help**	12 national helplines + plain-language glossary

3.1 Core innovation — explainable eligibility

Most "scheme finders" return a bare list. Bharat Setu returns a **verdict with evidence**. For every scheme the engine emits structured reasons and blockers, e.g.:

> *✓ Age 34 is within the eligible band 18–120 ✓ Rural residence matches the scheme area ✓ Land holding 1.5 acre(s) is within limits ✕ Income limit is ₹6,00,000/year*

This is produced by a **rule engine over a machine-readable `rules{}` block** stored with each scheme — not by an LLM. That means it is deterministic, auditable, instantaneous, and works offline.

3.2 Core innovation — grounded AI with graceful degradation

The optional LLM layer is **never the source of truth**. The serverless function builds its system prompt directly from the same curated knowledge base the UI uses and is instructed to cite only official `.gov.in` / `.nic.in` portals. If no key is configured, the upstream is down, or the response looks

ungrounded, the handler returns `{ok:false, fallback:true}` and the client seamlessly answers from the **deterministic offline engine**. The user therefore never sees a fabricated scheme, amount or URL — the failure mode is *less rich*, never *wrong*.

3.3 Core innovation — offline-first

All eligibility, document, letter and fraud logic ships as static JavaScript and is cached by a **service worker**. A citizen in a low-connectivity district retains full functionality. This is the single most important design decision for the "Bharat" context.

3.4 Designed for the actual user

- **13 languages** — English, Hindi, Bengali, Tamil, Telugu, Marathi, Gujarati, Kannada,

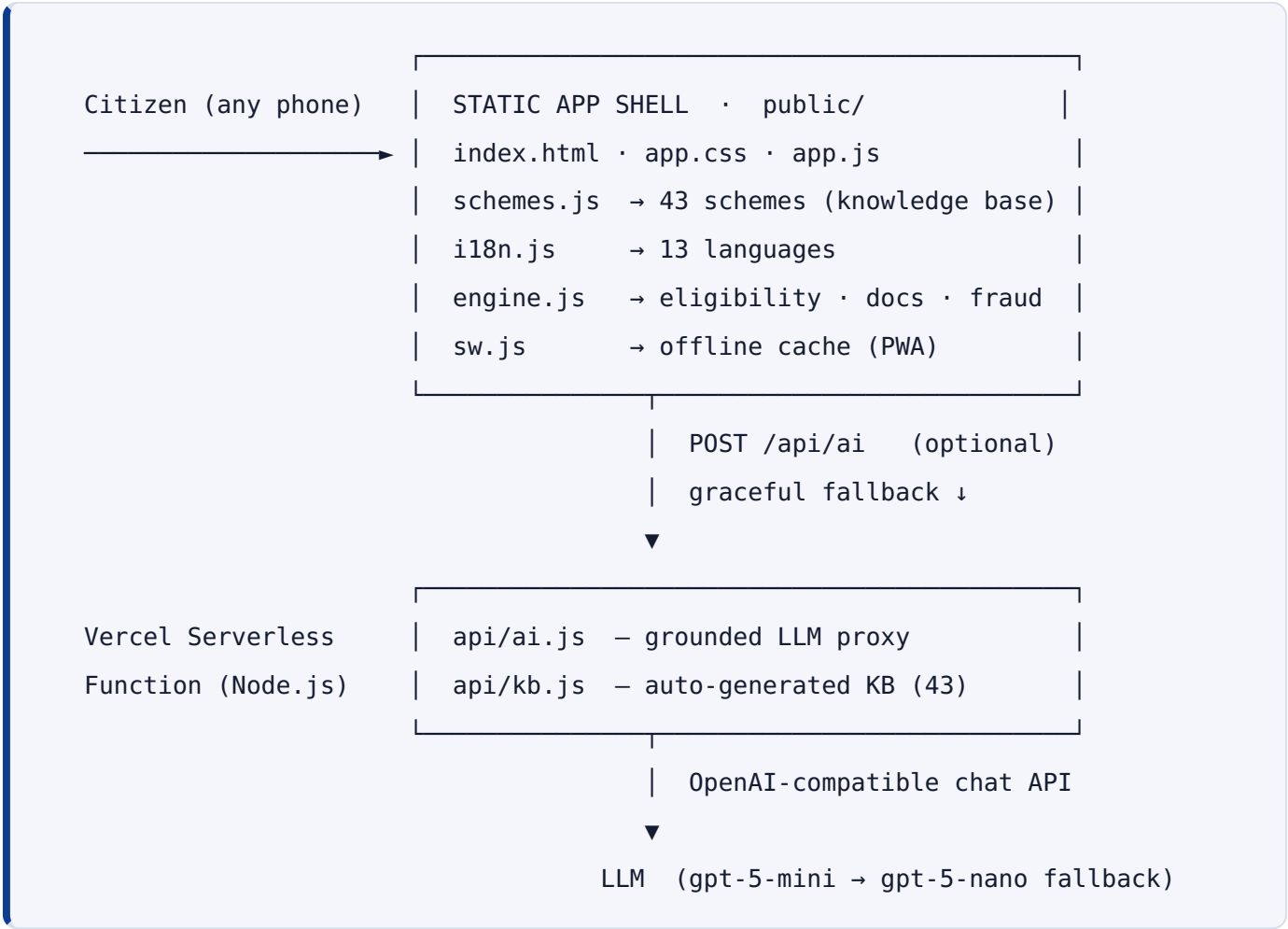
Malayalam, Punjabi, Odia, Assamese, Urdu (with full RTL layout).

- **Voice in & out** via the Web Speech API, locale-matched per language.
- **Font scaling to 140 %**, **high-contrast mode**, **dark mode**.
- **Simple Mode** — one tap enlarges text, calms the layout and hides decorative elements for

first-time internet users.

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4. System Architecture



4.1 Deliberate architectural choice — zero build step

The application is a **plain static bundle plus one serverless function**. There is no bundler, no front-end framework and no runtime filesystem access. Consequences:

1. The **same `api/ai.js` handler** runs under the local dev server *and* on Vercel, so local

and production behaviour are identical.

1. A static bundle is trivially cacheable, and caching is what makes offline mode viable.
2. Tiny footprint → fast on 3G and on low-end devices, which is the target.

4.2 Technology stack

Layer	Technology	Rationale
Front end	Semantic HTML5, hand-written CSS design system, **vanilla JS**	No framework cost; runs on any device
Reasoning	Custom rule engine + NLU + fraud scoring in `engine.js`	Deterministic, offline, auditable
Back end	**One Vercel serverless function** (`api/ai.js`)	Minimum surface, Vercel-native
AI	Any OpenAI-compatible chat endpoint (`AI_MODEL`, `AI_MODEL_FALLBACK`)	Provider-agnostic, swappable
Offline	Service Worker + Web App Manifest (PWA)	Works without network
Voice	Web Speech API (`SpeechRecognition`, `speechSynthesis`)	Free, on-device, 13 locales
Hosting	**Vercel** (static + functions) · **GitHub** (source)	One-click, free tier

4.3 Repository layout

```
webapp/
├── api/
│   ├── ai.js                # serverless AI proxy (chat | letter | fraud)
│   └── kb.js                # auto-generated grounded knowledge base
├── public/
│   ├── index.html           # 8-view app shell
│   ├── manifest.webmanifest  # PWA manifest
│   ├── sw.js                # offline service worker
│   └── static/
│       ├── app.css          # design system
│       ├── app.js           # UI shell, views, My Plan tracker
│       ├── schemes.js       # 43 real schemes (the knowledge base)
│       ├── i18n.js          # 13 languages
│       ├── engine.js        # offline reasoning engine
│       └── icon.svg / icon-192.png / icon-512.png
├── scripts/
│   ├── dev-server.mjs       # local + sandbox server (Vercel-parity)
│   ├── build.mjs            # build validator (no bundler needed)
│   ├── verify.mjs           # 63-test end-to-end suite
│   ├── generate-kb.mjs      # schemes.js → api/kb.js
│   ├── make-icons.mjs       # zero-dependency PNG icon generator
│   └── add-plan-i18n.mjs    # multilingual string injector
├── vercel.json              # Vercel config (build, functions, headers)
├── ecosystem.config.cjs     # PM2 config for the sandbox
└── package.json
```

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5. The Knowledge Base

43 real Government of India / State schemes, each with a machine-readable rule block, official portal, required documents and application steps. Coverage spans **30 categories**:

`farmer · cash · pension · worker · bpl · food · family · women · energy · rural · water · housing · urban · health · all · girls · savings · education · disability`

`· senior · student · loan · skill · employment · business · scst · insurance · safety · rights · governance · document`

Representative schemes: **PM-Kisan**, **PM-JAY / Ayushman Bharat**, **PM Awas Yojana (Gramin & Urban)**, **MGNREGA**, **NFSA**, **NSAP pensions**, **Sukanya Samriddhi**, **Atal Pension Yojana**, **Mudra**, **PM SVANidhi**, **Ujjwala**, **Jal Jeevan Mission**, **RTE**, **PM Vishwakarma**, **e-Shram**, **UDID**, **DigiLocker**, **Passport Seva**, **Ni-kshay (TB)**, and **RTI**.

The knowledge base also powers a **catalogue of 83 distinct documents**, each with a guide stating *who issues it* and *how long it takes*, plus **12 national helplines**.

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6. Feature Detail and Walkthrough

6.1 Scheme Finder — explainable matching

The engine evaluates hard identity conditions (women-only, widow, disability, TB, girl-child, farmer, employed …) as blockers, then scores soft/quantitative conditions (age band, income ceiling, land holding, rural/urban, category interest). Verdicts: `high` / `mid` / `low` / `none`. Every surviving match carries human-readable reasons; every blocked scheme shows why.

6.2 What-if simulator — eligibility as a dial, not a verdict (new)

Eligibility feels fixed and final to a citizen, but it is really a function of a few numbers. The simulator puts **age** and **annual income** on live sliders and re-runs the full rule engine on every move — without touching the saved profile. As the sliders move, the citizen watches three things change in real time:

- the **number of matched schemes** and the count of strong matches;
- the **estimated annual benefit value**;
- a diff against their current result: **"Newly eligible"** and **"Would no longer qualify"**.

This turns a static list into a *planning* tool. A parent can see exactly how far over an income ceiling they are; a young person can see what opens up at a given age. It runs entirely offline.

6.3 Household mode — one plan for the whole family (new)

Welfare in India is claimed by households, not by individuals. Household mode lets the user **save several family members**, each with their own full profile, then:

- switch between members with one tap (loading a member re-runs their eligibility instantly);
- view a **household summary table** — each member's match count, strong-match count, top three

schemes and estimated annual value — with a **combined household total**;

- **export the whole household plan** as a single text file to carry to a CSC or Panchayat office.

Member profiles are stored only in the browser's `localStorage`. Nothing is uploaded.

6.4 My Plan — from discovery to disbursement (new)

Discovery is only the first mile. My Plan turns a match list into an actionable plan:

- **Application tracker** — mark each scheme *Not started → Applied → Benefit received →

Rejected (escalate)*, with an optional reference-number note.

- **Document-readiness bar** per scheme, driven by the Document Checker state.
- **Estimated annual benefit value in ₹**, computed conservatively by parsing each matched

scheme's headline amount (monthly amounts annualised). Presented as an itemised table with a total — a tangible, motivating number.

- **Export** the whole plan as a `.txt`, **share** it via the Web Share API, or **print** it to

PDF to carry to a CSC, bank or Gram Panchayat office.

6.5 AI Assistant — grounded, multilingual

Text or voice in any of 13 languages. The classifier detects intent (eligibility, documents, how-to, status, amount, grievance, RTI, fraud, health, education, employment, pension, agriculture, housing, women, water, help), extracts entities (age, income, land) and selects evidence schemes. The LLM layer refines the answer; the offline engine guarantees one always exists.

6.6 Document Checker

Tick what you already have → a readiness percentage, the exact missing documents, and for each one the **issuing office** and **expected timeline** (e.g. *Income certificate — apply on the state e-District portal or CSC; the Tehsildar/Revenue office issues it; 7–15 days*).

6.7 RTI & Letters

Four legally-shaped templates — **RTI application** (cites Section 6(1), the ₹10 fee, the 30-day statutory deadline and the appellate authority), **public grievance**, **scheme representation**, and **certificate request**. Each is refined by the AI when available and always available offline. Print, copy or download.

6.8 Fraud Shield

A transparent, weighted rule set scores a pasted SMS/WhatsApp message out of 100: OTP/PIN requests, artificial urgency, shortened/throwaway domains, up-front "processing fees", lottery claims, APK/screen-share pushes, bank-detail harvesting, brand impersonation, job scams, fake free-scheme offers, and generic mass-mailing language. Genuine `.gov.in` / `.nic.in` domains register as a *positive* signal (with a negative lookahead so a phishing lookalike such as `pmkisan.gov.in.xyz` cannot masquerade as safe). Output: verdict, the red flags explained, and concrete next steps (call **1930** within the golden hour, report on cybercrime.gov.in, block the card, report to Sanchar Saathi).

6.9 Accessibility and inclusion

Skip link, ARIA live regions, keyboard-operable controls, semantic landmarks, 13 languages with correct `dir="rtl"` for Urdu, font scaling to 140 %, high-contrast and dark themes, optional spoken output for every answer, and a dedicated **Simple Mode** for first-time users.

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7. Deployment and Verification

7.1 Live deployment

Item	Value
Production URL	https://webapp-kohl-six-95.vercel.app
Platform	Vercel (static output `public/` + one serverless function)
Source	https://github.com/no6546559-sketch/bharat-setu (branch `main`)
Status	✅ Live, HTTP 200, public

7.2 API surface

Method	Path	Purpose
`GET`	`/api/ai`	Status: `{ ok, ai_enabled, model, schemes, modes }`
`POST`	`/api/ai`	`mode:"chat"` grounded Q&A · `mode:"letter"` letter drafting · `mode:"fraud"` second-opinion fraud analysis

7.3 Automated verification

`node scripts/verify.mjs` executes **63 assertions against the shipped bytes** (not a re-implementation), covering:

- **Eligibility engine** — ranking, farmer/BPL/women paths, hard exclusions, age boundaries, and income ceilings (including the corrected ESIC ₹3,00,000 limit case).
- **Document intelligence** — catalogue build, readiness scoring, per-document guides.
- **Fraud shield** — scam detection, official-domain positive signal.
- **Multilingual NLU** — intent classification and entity extraction (age/income/land).
- **Grounded offline answers** — citations, RTI routing, fraud routing.
- **Letter generator** — all four letter types.
- **i18n completeness** — all 13 languages.
- **Live `/api/ai` endpoint** — status and graceful fallback.

Result: 63 passed, 0 failed. The build validator additionally confirms every scheme carries an official portal link, all scheme IDs are unique, and both serverless modules are present.

7.4 Graceful degradation verified in production

On the live deployment the API currently reports `ai_enabled:false` (no LLM key configured in the Vercel project) and `POST /api/ai` returns `{ok:false, fallback:true}`. The client transparently serves the deterministic offline engine — **every feature remains fully functional**. This is the intended failure mode and is validated by the test suite.

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8. Mapping to Evaluation Criteria

Criterion	How Bharat Setu satisfies it
Innovation & Originality	Explainable rule-based eligibility (verdict *with evidence*) + grounded AI that degrades to a deterministic engine; a **What-if simulator** that turns eligibility into a live dial; **Household mode** that plans for the whole family; ₹ benefit estimator; document-intelligence with issuing office and timelines; Fraud Shield that explains *why*
Technical Implementation	43-scheme machine-readable KB, custom NLU, 13-language i18n, PWA offline, service worker, serverless AI proxy with rate limiting and model fallback, 63-test suite, build validator, zero-dependency icon generator, generated PPTX deck
Real-World Impact	Directly addresses the last-mile information gap in India's welfare delivery; targets the exact groups most excluded (rural, women, seniors, disabled, first-time users); household-level planning matches how welfare is actually claimed
Usability	13 languages incl. RTL, voice in/out, font scaling to 140 %, high contrast, dark mode, Simple Mode, plain-language glossary, one-tap demo profile, live ₹ value pill, slider-based exploration
Scalability	Fully static front end on a global CDN; stateless serverless function; KB is a data file — adding schemes/languages is additive; no database to scale
Presentation & Demo	Live URL + GitHub + this report + a 14-slide deck; demo path: sample profile → ranked schemes → What-if sliders → Household summary → My Plan ₹ estimate → document readiness → RTI letter → fraud check

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9. Limitations and Responsible Use

- Benefit amounts and eligibility criteria are **indicative** and change by government

notification. Every scheme card links the official portal and the app instructs the user to confirm before applying.

- This is **not a government website**. It does not accept applications, hold funds, or

represent any department.

- **Privacy:** all profile data, tracker state and chat history live only in the browser's

`localStorage`. Nothing is uploaded. Only free-text the user explicitly submits (question, letter details, suspicious message) is forwarded to the LLM — and only while online.

- The app never requests, and should never be given, OTPs, PINs, CVVs or full bank/Aadhaar numbers.

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10. Future Roadmap

1. **More schemes and state-specific top-ups** (currently pan-India plus popular state schemes).
2. **DigiLocker integration** to auto-verify documents and pre-fill the checklist.
3. **SMS/IVR channel** for feature phones, extending reach beyond smartphones.
4. **On-device OCR** to read a photograph of a government notice.
5. **Status polling** via official portals where an API exists, to auto-update My Plan.
6. **Offline-first localisation** of scheme *descriptions* (currently hero text + scheme names/ benefits are localised, with AI handling long-form explanation).

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11. Conclusion

Bharat Setu demonstrates that the hardest problem in Indian welfare delivery is not eligibility — it is **access**. By combining an **explainable, offline, multilingual reasoning engine** with a **grounded AI layer that can only ever add richness, never risk correctness**, the project delivers a tool that a first-time internet user in a low-connectivity district can genuinely use to find, claim and track the benefits they are owed.

It is live, open-source, fully tested, and built to run where it is needed most — at the edge, on a phone, in the citizen's own language.

Built as an independent citizen-help tool for a hackathon. **Helplines:** 🚨 112 emergency · 👩 181 women · 💳 1930 cyber fraud · 🧒 1098 child

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